



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023-2024)

TC: @Chalmeram

MINOR
23-04-2023

PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE : MEC-I (B)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

Test Booklet Code

This Booklet contains 28 pages.

E2

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
- The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject. **Candidates are advised to read all 15 questions in each subject of Section B** before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
- Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one** mark will be deducted from the total scores. **The maximum marks are 720.**
- Use **Blue/Black Ball Point Pen only** for writing particulars on this page/mark responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
- No candidate, without special permission of the Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

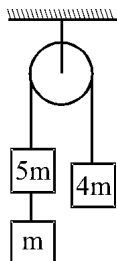
Candidate's Signature : _____ Invigilator's Signature : _____

TC: @Chalmeram

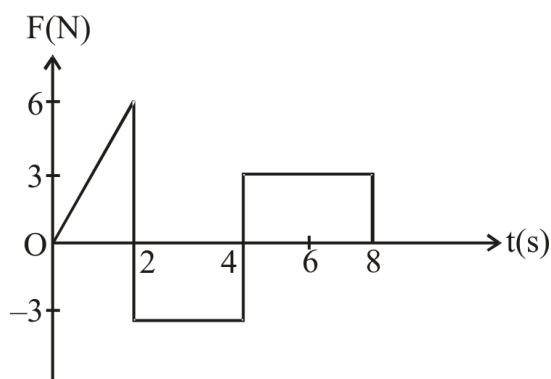
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SECTION-A (PHYSICS)

1. For the figure shown what will be the tension in the rope connecting blocks of mass m and $5m$:-



- (1) $\frac{4}{5}mg$ (2) $\frac{1}{5}mg$
(3) $\frac{6}{5}mg$ (4) Zero
2. The force F acting on a particle of mass M is indicated by the force-time graph shown below. The change in momentum of the particle over the time interval from 0 to 8 seconds is :



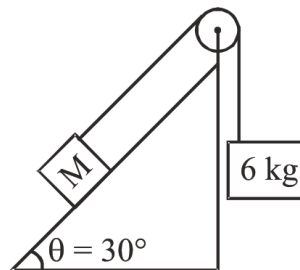
- (1) 24 Ns (2) 20 Ns (3) 12 Ns (4) 6 Ns
3. Three blocks A, B and C of masses 4 kg, 2 kg and 1 kg respectively, are in contact on a frictionless surface, as shown. If a force of 14 N is applied on the 4 kg block, then the contact force between A and B is :



- (1) 6 N
(2) 8 N
(3) 18 N
(4) 2 N

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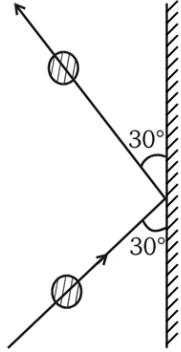
4. Find the value of ' M ' in the given figure, if system remains in equilibrium :-



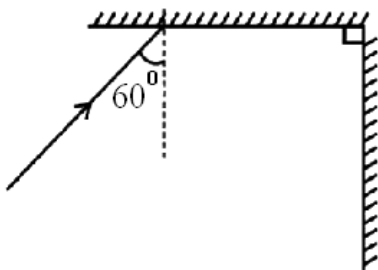
- (1) 6 kg (2) 12 kg (3) 24 kg (4) 4 kg
5. A body whose momentum is constant, must have constant :-
(1) Force (2) Velocity
(3) Acceleration (4) All of these
6. The coefficient of static friction between two surfaces depend on :-
(1) Nature of surface
(2) Area of contact
(3) Shape of surface in contact
(4) None
7. Four forces act on a point object. The object will be in equilibrium if :-
(1) They are opposite to each other in pairs.
(2) The sum of x, y and z components of forces is zero separately.
(3) They form a closed figure of 4 sides when added as per polygon law.
(4) All of above
8. A lift is moving down with an acceleration " a ". A man in the lift drops a ball inside the lift, find acceleration of the ball with respect to man in the lift ? (Take down direction positive)

- (1) a (2) $(g - a)$
(3) g (4) $g + a$

9. A 0.5 kg ball moving with a speed of 12 m/s strikes a hard wall at an angle of 30° with the wall. It is reflected with the same speed and at the same angle. If the ball is in contact with the wall for 0.25 seconds, the average force acting on the wall is :-

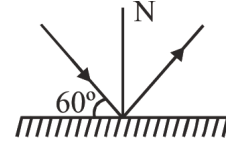


- (1) 48 N (2) 24 N
(3) 12 N (4) 96 N
10. In vacuum,
- (1) red travels faster.
(2) yellow travels faster.
(3) violet travels faster.
(4) all colours have same speed.
11. A plane mirror makes an angle of 30° with the horizontal. If a vertical ray strikes the mirror, find the angle between the mirror and the reflected ray ?
- (1) 30° (2) 90° (3) 60° (4) 40°
12. Two plane mirrors are arranged at right angles to each other as shown in figure. A ray of light is incident on horizontal mirror at an angle of incidence 60° . The total deviation of the ray after reflection from both mirror will be :-

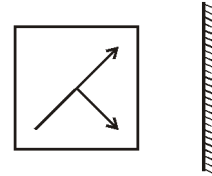


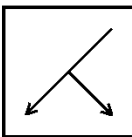
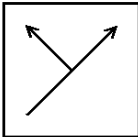
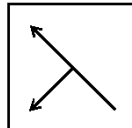
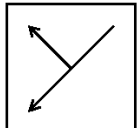
- (1) 60° (2) 120° (3) Zero (4) 180°

13. Find the angle of reflection :

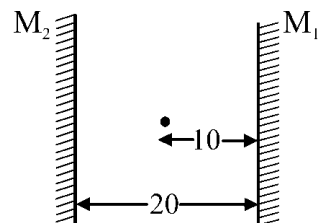


- (1) 30° (2) 60°
(3) 70° (4) None of these
14. It is desired to photograph the image of an object placed at a distance of 3 m from a plane mirror. The camera, which is at a distance of 4.5 m from the mirror should be focused for a distance of -
- (1) 3 m (2) 4.5 m (3) 6 m (4) 7.5 m
15. Choose the correct mirror-image of figure given below.



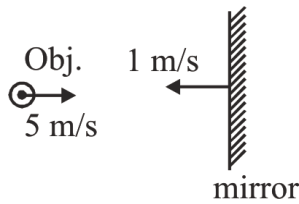
- (1)  (2) 
(3)  (4) 

16. A ray of light is incident on a plane mirror at an angle ' θ '. If deviation is 120° then θ will be :
- (1) 120° (2) 90° (3) 60° (4) 30°
17. Two plane mirrors M_1 & M_2 are placed as shown and an object is placed in b/w them at a distance of 10cm from M_1 . If separation between mirror is 20cm then distance of first two images from M_2 will be (in cm) :-



- (1) 30, 60 (2) 10, 30 (3) 20, 30 (4) 10, 20

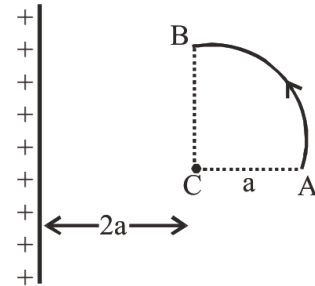
18. An object moves with 5 m/s towards right while the mirror moves with 1 m/s towards the left as shown obtained the velocity of image.



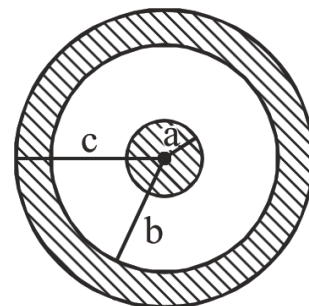
- (1) 3 m/s towards right
(2) 3 m/s towards left
(3) 7 m/s towards right
(4) 7 m/s towards left
19. A uniform electric field exists in x-y plane. The potential of points A(2m, 2m), B(-2m, 2m) and C(2m, 4m) are 4V, 16V and 12V respectively. The electric field is :
- (1) $(4\hat{i} + 5\hat{j})$ V/m
(2) $(3\hat{i} + 4\hat{j})$ V/m
(3) $(3\hat{i} - 4\hat{j})$ V/m
(4) $(3\hat{i} - 4\hat{j})$ V/m
20. Surface charge density of a thin disc having radius R varies with distance r from centre as $\sigma = \sigma_0 \frac{R}{r}$, ($r \neq 0$) then total charge of disc is.
- (1) $2\pi\sigma_0 R^3$
(2) $\sqrt{2}\pi\sigma_0 R^2$
(3) $2\sqrt{2}\pi\sigma_0 R^2$
(4) $2\pi\sigma_0 R^2$
21. The potential function of electrostatic field is given by $V = 2x^2 - 3xy^2z + 4yz$ then electric field at point (0, 1, 1)m will be
- (1) $3\hat{i} - 4\hat{j} + 4\hat{k}$
(2) $2\hat{i} - 4\hat{k}$
(3) $3\hat{i} - 4\hat{j} - 4\hat{k}$
(4) $-3\hat{i} + 4\hat{j} + 4\hat{k}$

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22. The arc AB with the centre C and the infinitely long wire having linear charge density λ are lying in the same plane. The amount of work to be done to move a point charge q_0 from point A to B through a circular path AB of radius a is equal to:-

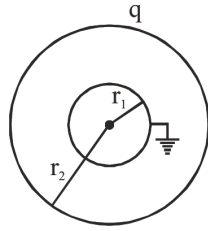


- (1) $\frac{q_0\lambda}{4\pi\epsilon_0} \ln\left(\frac{2}{3}\right)$
(2) $\frac{q_0\lambda}{2\pi\epsilon_0} \ln\left(\frac{3}{2}\right)$
(3) $\frac{q_0\lambda}{2\pi\epsilon_0} \ln\left(\frac{2}{3}\right)$
(4) None of these
23. A solid conducting sphere of radius a has a net positive charge 2Q. A conducting spherical shell of inner radius b and outer radius c is concentric with the solid sphere and has a net charge -Q. The surface charge density on the inner and outer surfaces of the spherical shell will be :-

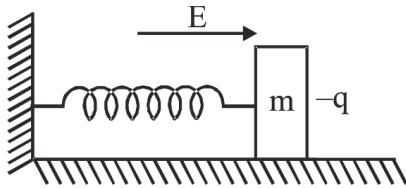


- (1) $-\frac{2Q}{4\pi b^2}, \frac{Q}{4\pi c^2}$
(2) $-\frac{Q}{4\pi b^2}, \frac{Q}{4\pi c^2}$
(3) $0, \frac{Q}{4\pi c^2}$
(4) None of the above

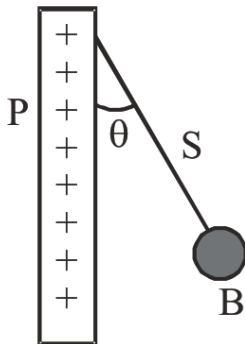
24. Two conducting thin shells of radii r_1 and r_2 are placed as shown. The outer sphere is given a charge q . The charge x on inner sphere will be (inner sphere is grounded).



- (1) q (2) $-q$
 (3) $\frac{-qr_1}{r_2}$ (4) $\frac{-qr_2}{r_1}$
25. A block of mass m containing a net negative charge $-q$ is placed on a frictionless horizontal table and is connected to a wall through an unstretched spring of spring constant k . If the horizontal electric field E parallel to the spring is switched on, then the maximum compression of the spring is :-



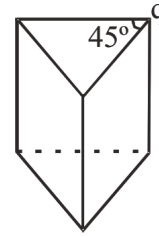
- (1) $\sqrt{qE/k}$ (2) $2qE/k$ (3) qE/k (4) zero
26. A charged ball B hangs from a silk thread S, which makes an angle θ with a large charged conducting sheet P, as shown in the figure. The surface charge density σ of the sheet is proportional to :-



- (1) $\sin \theta$ (2) $\tan \theta$
 (3) $\cos \theta$ (4) $\cot \theta$

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27. A prism shaped imaginary structure is given. A point charge is kept as given in figure. Calculate electric flux passing through the prism.

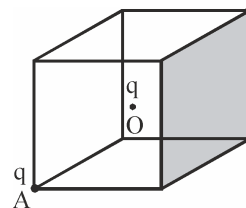


- (1) $\frac{q}{4\epsilon_0}$
 (2) $\frac{q}{8\epsilon_0}$
 (3) $\frac{q}{16\epsilon_0}$
 (4) $\frac{q}{32\epsilon_0}$
28. A charged particle q is shot towards another charged particle Q which is fixed, with a speed v . It approaches Q upto a closest distance r and then returns. If q were given a speed $2v$, the closest distance of approach would be :-



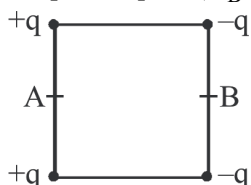
- (1) r (2) $2r$
 (3) $r/2$ (4) $r/4$

29. Two point charge, each of q , are kept at centre of cube O and vertex A. Find ϕ_e from shaded face.



- (1) $\frac{5q}{8\epsilon_0}$
 (2) $\frac{7}{24} \frac{q}{\epsilon_0}$
 (3) $\frac{2q}{\epsilon_0}$
 (4) $\frac{5}{24} \frac{q}{\epsilon_0}$

30. Four electric charges $+q$, $+q$, $-q$ and $-q$ are placed at the corners of a square of side $2L$ as shown in figure. The electric potential difference between point A, midway between the two charges $+q$ and $+q$, and point B midway between the two charges $-q$ and $-q$ is : ($V_B - V_A$)

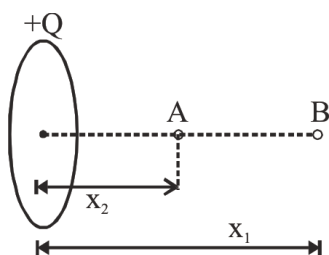


- (1) $\frac{1}{4\pi\epsilon_0} \frac{4q}{L} \left(\frac{1}{\sqrt{5}} - 1 \right)$
 (2) $\frac{1}{4\pi\epsilon_0} \frac{2q}{L} \left(1 + \frac{1}{\sqrt{5}} \right)$
 (3) $\frac{1}{4\pi\epsilon_0} \frac{2q}{L} \left(1 - \frac{1}{\sqrt{5}} \right)$
 (4) zero
31. If linear charge density of a wire of length L depends on distance x from one end as $\lambda = \frac{\lambda_0(x)}{L}$.



Find total charge of wire.

- (1) $\lambda_0 L$ (2) $\frac{\lambda_0 L}{2}$
 (3) $\frac{\lambda_0 L}{4}$ (4) None of these
32. Electric field due to charged ring on axis at a point A very close to centre is same as at point B far away from centre. The radius of ring is :-



- (1) $\sqrt{x_1 x_2}$ (2) $(x_1^2 x_2)^{1/3}$
 (3) $(x_2^2 x_1)^{1/3}$ (4) $x_1 + x_2$

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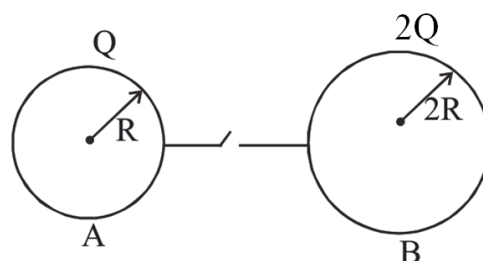
33. Charges $+q$, $-q$, $+q$ and $-q$ are placed at the corners A, B, C and D respectively of a square of side a . The potential energy of the system is $\frac{1}{4\pi\epsilon_0}$ times :-

- (1) $\left(\frac{q^2}{a} \right) (-4 + \sqrt{2})$
 (2) $\left(\frac{q^2}{2a} \right) (-4 + \sqrt{2})$
 (3) $\left(\frac{4q^2}{a} \right)$
 (4) $-\left(\frac{4\sqrt{2}q^2}{a} \right)$

34. An insulated sphere of radius R has a uniform volume charges density ρ . The electric field at a point P inside the sphere at a distance r from the centre is

- (1) $\frac{R\rho}{3\epsilon_0}$
 (2) $\frac{r\rho}{3\epsilon_0}$
 (3) Zero
 (4) $\frac{2}{3} \left(\frac{r\rho}{3\epsilon_0} \right)$

35. Two conducting spheres A and B having radius R & $2R$ and charge Q & $2Q$ respectively, if they are connected by wire then :-



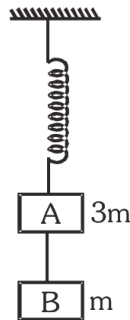
- (1) No charge will flow through the wire
 (2) $Q/3$ charge will flow from B to A
 (3) $Q/3$ charge will flow from A to B
 (4) $2Q/3$ charge will flow from B to A

SECTION-B (PHYSICS)

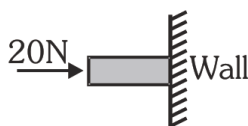
36. A block A of mass m_1 rests on a horizontal table. A light string connected to it passes over a frictionless pulley at the edge of table and from its other end another block B of mass m_2 is suspended. The coefficient of kinetic friction between the block and the table is μ_k . When the block A is sliding on the table, the tension in the string is :-

(1) $\frac{(m_2 - \mu_k m_1)g}{(m_1 + m_2)}$ (2) $\frac{m_1 m_2 (1 + \mu_k)g}{(m_1 + m_2)}$
 (3) $\frac{m_1 m_2 (1 - \mu_k)g}{(m_1 + m_2)}$ (4) $\frac{(m_2 + \mu_k m_1)g}{(m_1 + m_2)}$

37. Two blocks A and B of masses $3m$ and m respectively are connected by a massless and inextensible string. The whole system is suspended by a massless spring as shown in figure. The magnitudes of acceleration of A and B immediately after the string is cut, are respectively :-



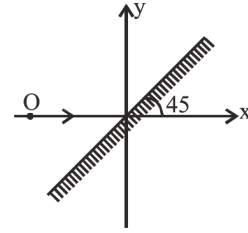
- (1) $\frac{g}{3}, g$ (2) g, g
 (3) $\frac{g}{3}, \frac{g}{3}$ (4) $g, \frac{g}{3}$
38. A horizontal force of 20 N is necessary to just hold a block stationary against a wall. The coefficient of friction between the block and wall is 0.15. The weight of the block is :-



- (1) 20 N (2) 3 N
 (3) 30 N (4) 0.3 N

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39. Image of the object can be :-

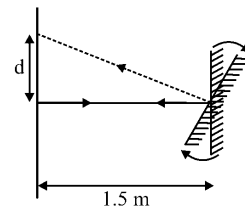


- (1) $\downarrow$ (2) $\uparrow$ (3) $\rightarrow$ (4) $\nwarrow$

40. When a ray of light is incident normally on a plane mirror then the angle of reflection will be :-

- (1) 0° (2) 90° (3) 180° (4) -90°

41. Light incident normally on a plane mirror attached to a galvanometer coil retraces backwards as shown in figure. A current in the coil produces a deflection of 3.5° of the mirror. What is the displacement of reflected spot of light on a screen placed 1.5 m away ?

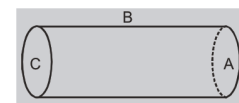


- (1) 10.5 cm (2) 18.31 cm
 (3) 15 cm (4) 3 cm

42. A plane mirror is approaching an object at a speed of 10 cm/sec. If distance between object & mirror is 50 cm then find distance between object & its image in mirror after 3 second.

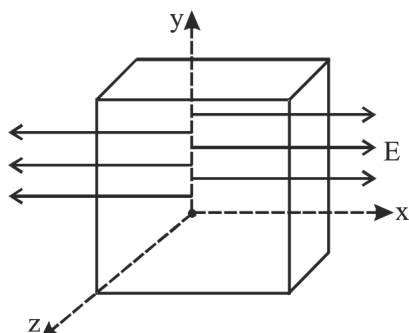
- (1) 20 cm (2) 10 cm
 (3) 40 cm (4) None of these

43. A hollow cylinder has a charge q coulombs located within it symmetrically. If ϕ is the electric flux in units of volt-meter associated with the curved surface B, the flux linked with the plane surface A in units of volt-meters will be :-

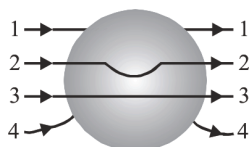


- (1) $\frac{q}{\epsilon_0} - \phi$ (2) $\frac{1}{2} \left[\frac{q}{\epsilon_0} - \phi \right]$
 (3) $\frac{q}{2\epsilon_0}$ (4) $\frac{\phi}{3}$

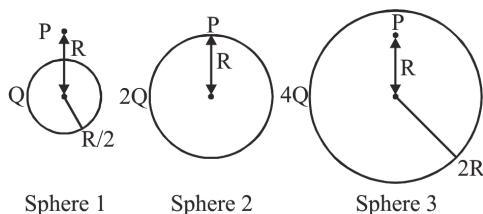
44. Electric field at $x = 10$ cm is 100 V/m and at $x = -10$ cm is -100 V/m. The magnitude of charge enclosed by the cube of side 20 cm centred at origin:-



- (1) $8 \epsilon_0$ (2) $2 \epsilon_0$
(3) $3 \epsilon_0$ (4) $5 \epsilon_0$
45. A metallic solid sphere is placed in a uniform electric field then lines of force follow the path (s) shown in figure as



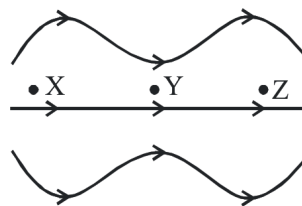
- (1) 1 (2) 2
(3) 3 (4) 4
46. Charges Q , $2Q$ and $4Q$ are uniformly distributed in three dielectric solid spheres 1, 2 and 3 of radii $R/2$, R and $2R$ respectively, as shown in figure. If magnitudes of the electric fields at point P at a distance R from the centre of spheres 1, 2 and 3 are E_1 , E_2 and E_3 respectively, then :-



- (1) $E_1 > E_2 > E_3$
(2) $E_3 > E_1 > E_2$
(3) $E_2 > E_1 > E_3$
(4) $E_3 > E_2 > E_1$

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47. In this figure, electric field lines in a certain region are shown. The figure suggests that



- (1) $E_x = E_y = E_z$ (2) $E_x < E_y < E_z$
(3) $E_x > E_y > E_z$ (4) $E_x = E_z < E_y$
48. Consider the following statements about electric dipole and select the correct ones :-
S1 : Electric dipole moment vector $\vec{p}$ is directed from negative charge to positive charge.
S2 : The electric field of a dipole at a point with position vector $\vec{r}$ depends on $|\vec{r}|$ as well as the angle between $\vec{r}$ and $\vec{p}$.
S3 : The electric dipole potential falls off as $\frac{1}{r^2}$ and not as $\frac{1}{r}$.
S4 : In a uniform electric field, the electric dipole experiences no net forces but a torque $\vec{\tau} = \vec{p} \times \vec{E}$.

- (1) S2, S3 and S4 (2) S3 and S4
(3) S2 and S3 (4) all four
49. An electric dipole is placed in a non uniform electric field, then it experiences :-
(1) a force which must be zero
(2) a torque which must be non zero
(3) a force which must be non zero
(4) both a force and a torque which may be non zero
50. The charges each Q are placed at $x = d, 2d, 4d, \dots, \infty$. Find electric potential at origin O :-

- (1) $\frac{4kd}{d}$ (2) $\frac{3kd}{d}$ (3) $\frac{2kQ}{d}$ (4) $\frac{kQ}{d}$

Topic : CONFORMATIONAL (FROM BUTANE), OPTICAL ISOMERISM, *TG: @Chalnaayaaar* GOC-I, SOLUTION, ELECTROCHEMISTRY (UPTO GALVANIC CELL)

SECTION-A (CHEMISTRY)

51. Select correct statement about electrophile.

- (1) Electrophiles are lewis bases
- (2) Electrophiles are electron deficient species
- (3) Electrophile have tendency to donate e^-
- (4) All of these

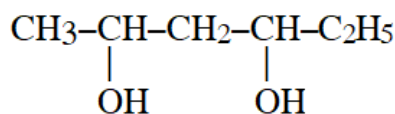
52. Select correct correct statement about carbocation ?

- (1) CH_3^+ is lewis base
- (2) CH_3^+ is paramagnetic in nature
- (3) CH_3^+ is sp^3 hybridised
- (4) CH_3^+ is electrophile

53. Which of the following has maximum +I effect ?

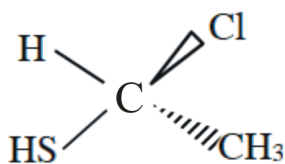
- (1) $-\text{CH}_3$
- (2) $\text{CH}_3-\text{C}(\text{CH}_3)_2$
- (3) $\text{CH}_3-\text{CH}(\text{CH}_3)-$
- (4) $\text{CH}_3-\text{CH}_2-\text{CH}_2-$

54. Calculate total number of optical active isomers of given compound.



- (1) 4
- (2) 3
- (3) 2
- (4) 1

55. Write down correct configuration.



- (1) S
- (2) R
- (3) E
- (4) Z

56. Select correct statement.

- (1) Meso compound is optically inactive
- (2) Racemic mixture is optically active mixture
- (3) Enantiomers are optically inactive
- (4) All are correct

57. Which of the cation is maximum stable cation.

- (1) $\text{CH}_3-\text{C}(\text{CH}_3)_2^+$
- (2) $\text{CH}_3-\text{CH}^+(\text{CH}_3)$
- (3) $\text{CH}_3-\text{CH}_2-\text{CH}_2^+$
- (4) $\text{CH}_3-\text{CH}_2^+$

58. Most stable ion is

- (1)
- (2)
- (3)
- (4)

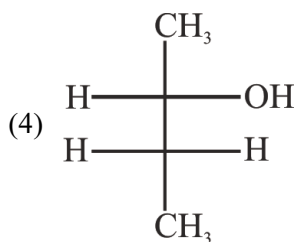
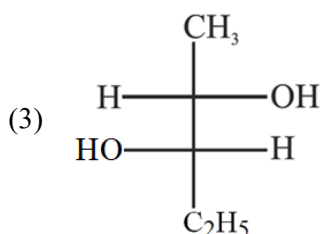
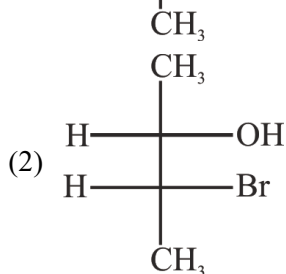
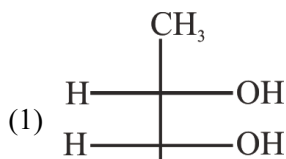
59. Correct order of -I effect is:

- (1) $-\text{N}^+\text{H}_3 > -\text{COOH} > -\text{NO}_2$
- (2) $-\text{N}^+\text{H}_3 > -\text{NO}_2 > -\text{COOH}$
- (3) $-\text{NO}_2 > -\text{COOH} > -\text{N}^+\text{H}_3$
- (4) $-\text{COOH} > -\text{NO}_2 > -\text{N}^+\text{H}_3$

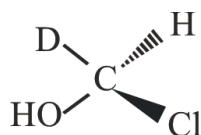
60. Correct stability order of conformer of cyclohexane is :-

- (1) Chair > Half-Chair > Boat > Twist-Boat
- (2) Chair > Half-Chair > Twist-Boat > Boat
- (3) Chair > Boat > Twist-Boat > Half-Chair
- (4) Chair > Twist-Boat > Boat > Half-Chair

61. Which of the following structure represents a threo stereoisomer :-

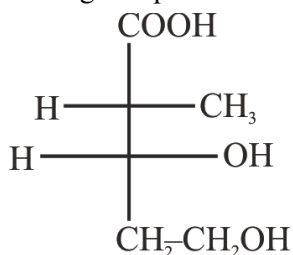


62. Correct absolute configuration of following molecule is :



- (1) R (2) S
(3) E (4) None

63. What are the configuration at the two chiral centres of following compound :



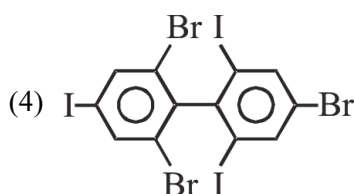
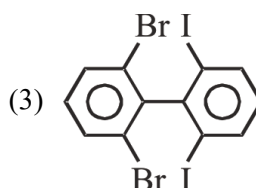
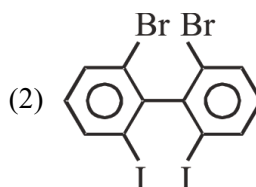
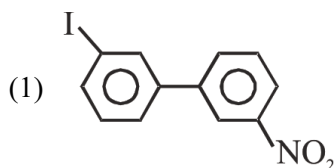
- (1) 2R, 3R (2) 2R, 3S
(3) 2S, 3R (4) 2S, 3S

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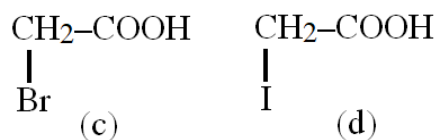
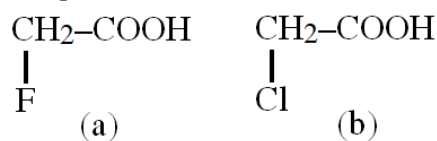
64. D-Glucose and L-Glucose are :-

- (1) Enantiomer
(2) Homomer
(3) Diastereomer
(4) Epimer

65. Which of the following biphenyls is optically active ?



66. Correct order of acidic strength of given compounds



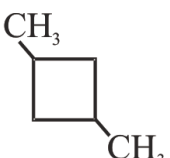
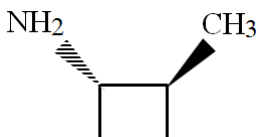
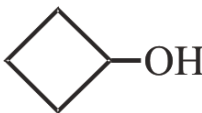
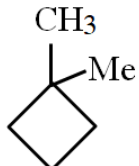
- (1) $d > c > b > a$
(2) $a > d > c > b$
(3) $a > b > c > d$
(4) $d > b > a > c$

67. **Statement-1** : In aqueous medium basic strength order is $(\text{CH}_3)_2\text{NH} > \text{CH}_3\text{-NH}_2 > (\text{CH}_3)_3\text{N}$

Statement-2 : In gaseous phase basic strength order is $(\text{CH}_3)_3\text{N} > (\text{CH}_3)_2\text{NH} > \text{CH}_3\text{-NH}_2$

- (1) Statement-1 is correct and statement-2 is incorrect
 (2) Statement-1 is incorrect and statement-2 is correct
 (3) Both statements are correct
 (4) Both statements are incorrect

68. Which of the following is optically active :-

- (1) 
- (2) 
- (3) 
- (4) 

69. Resistance and cell constant of a conductivity cell containing 0.01 mol/L NaOH solution is $5 \times 10^3 \Omega$ and 70 cm^{-1} respectively, then Λ_m of NaOH is -

- (1) $50 \text{ S-cm}^2 \text{ mol}^{-1}$
 (2) $100 \text{ S-cm}^2 \text{ mol}^{-1}$
 (3) $150 \text{ S-cm}^2 \text{ mol}^{-1}$
 (4) $1400 \text{ S-cm}^2 \text{ mol}^{-1}$

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70. Select the correct statement regarding a liquid solution formed by mixing carbon disulphide and acetone-

- (1) $\Delta V_{\text{mixing}} < 0$
 (2) $\Delta H_{\text{mixing}} = 0$
 (3) $\Delta S_{\text{mixing}} < 0$
 (4) Solution shows +ve deviation from Raoult's law.

71. K_H for solubility of N_2 gas in water is 10^5 atm at 298 K, then calculate number of moles of N_2 dissolved in 2 L water, at atmospheric pressure of 5 atm and 298 K, the mole fraction of N_2 in air is 0.8 :-

- (1) 5.55×10^{-3}
 (2) 4.44×10^{-3}
 (3) 2.22×10^{-2}
 (4) 1.11×10^{-2}

72. Osmotic pressure of decimolar solution of $\text{K}_4[\text{Fe}(\text{CN})_6]$ is 9.6 atm at 27°C , then what percentage of $\text{K}_4[\text{Fe}(\text{CN})_6]$ is unionised in solution : ($R = 0.08 \text{ atm L mol}^{-1} \text{ K}^{-1}$)

- (1) 25%
 (2) 50%
 (3) 75%
 (4) 90%

73. Van't Hoff factor (i) for trimerisation of a non volatile solute A in volatile solvent is (degree of association = 0.3)

- (1) 1.8
 (2) 0.8
 (3) 1.6
 (4) 1.2

74. Which of the following solution will have maximum equivalent conductance -

- (1) 0.1N KCl
- (2) 0.01 N KCl
- (3) 0.02 N KCl
- (4) 1 N KCl

75. Which colligative property is best suitable for determination of molar mass of biomolecules -

- (1) Relative lowering in vapour pressure
- (2) Elevation in boiling point
- (3) Depression in freezing point
- (4) Osmotic pressure

76. Calculate the amount of oxalic acid ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$) required to obtain 250 ml of semimolar solution.

- (1) 15.75 g
- (2) 1.575 g
- (3) 157.5 g
- (4) None

77. 6 gm of urea was dissolved in 9.9 moles of water. If the vapour pressure of pure water is P^0 , then vapour pressure of solution is :-

- (1) $0.1 P^0$
- (2) $1.1 P^0$
- (3) $0.9 P^0$
- (4) $0.99 P^0$

78. For an ideal solution, the correct option is :-

- (1) $\Delta_{\text{mix}} S = 0$
- (2) $\Delta_{\text{mix}} V \neq 0$
- (3) $\Delta_{\text{mix}} H = 0$
- (4) $\Delta_{\text{mix}} G = 0$

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79. The value of Henry's law constant for some gases at 293 K is given below. Arrange the gases in the increasing order of their solubility.

He : 144.97 kbar, H_2 : 69.16 kbar, N_2 : 76.48 kbar, O_2 : 34.86 kbar

- (1) $\text{He} < \text{N}_2 < \text{H}_2 < \text{O}_2$
- (2) $\text{O}_2 < \text{H}_2 < \text{N}_2 < \text{He}$
- (3) $\text{H}_2 < \text{N}_2 < \text{O}_2 < \text{He}$
- (4) $\text{He} < \text{O}_2 < \text{N}_2 < \text{H}_2$

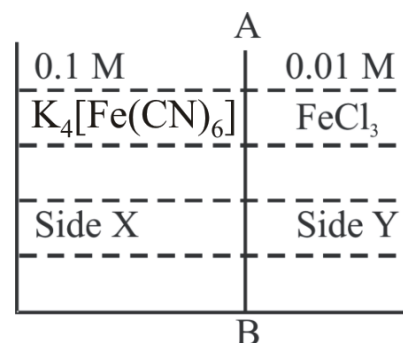
80. Which of the following aqueous solution will have the lowest freezing point ?

- (1) 0.10 m-sucrose
- (2) 0.10 m- NiCl_2
- (3) 0.10 m- CuSO_4
- (4) 0.10 m- NH_4NO_3

81. Mole fraction of vapours of A in mixture of A and B where $X_A = 0.4$ and $P_A^0 = 100$ mm of Hg, $P_B^0 = 200$ mm of Hg :

- (1) 0.4
- (2) 0.8
- (3) 0.25
- (4) 1.2

82. FeCl_3 on reaction with $\text{K}_4[\text{Fe}(\text{CN})_6]$ in aqueous solution give blue colour. These are separated by a semi-permeable membrane AB as shown. Due to osmosis, there is :-



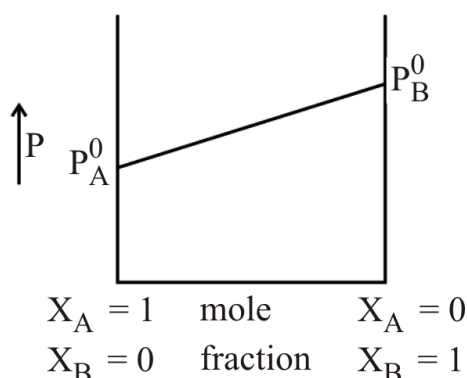
- (1) Blue colour formation in side X
- (2) Blue colour formation in side Y
- (3) Blue colour formation in both side X and Y.
- (4) No blue colour formation

SECTION-B (CHEMISTRY)

83. Which of the following statement is false ?

- (1) Raoult's law states that the vapour pressure of a component over a solution is proportional to its mole fraction in liquid phase.
- (2) Two sucrose solutions of same molality prepared in different solvents will have the same freezing point depression
- (3) Isotonic solutions have same osmotic pressure.
- (4) The osmotic pressure (π) of a solution is given by the equation $\pi = MRT$, where M is the molarity of the solution

84. The following is a graph plotted between the vapour pressure of two volatile liquid against their respective mole fractions, then correct statement is (are) :-



- (1) When $X_A = 1$ then $P = P_A^0$
- (2) When $X_B = 1$ then $P < P_A^0$
- (3) When $X_A = 1$ then $P < P_B^0$
- (4) Both (1) & (3)

85. A solute 'S' undergoes trimerization when dissolved in a certain solvent. The boiling point elevation of its 0.1 molal solution was found to be identical to the boiling point elevation in case of a 0.08 molal solution of another solute in same solvent which neither undergoes association nor dissociation. To what percent had the solute 'S' undergone trimerization ?

- (1) 30% (2) 40% (3) 50% (4) 60%

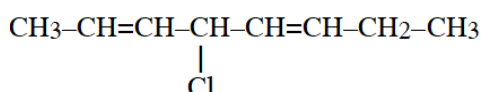
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86. Select correct statement(s).

- (a) Inductive effect is distance dependent.
- (b) Inductive effect of H atom is **zero**
- (c) Inductive effect is temporary effect.

- (1) a, b, c (2) b, c (3) a, c (4) a, b

87. Calculate total number of stereoisomers of given compound.

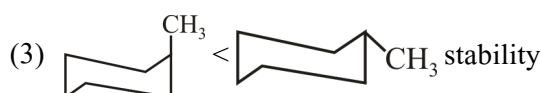
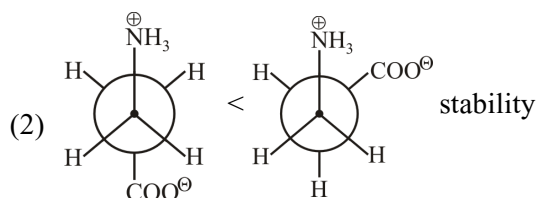
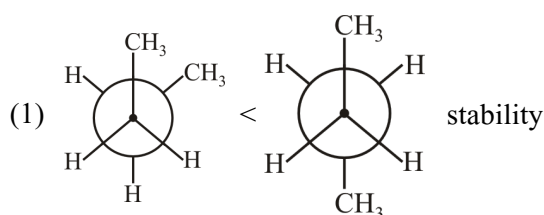


- (1) 8
- (2) 6
- (3) 4
- (4) 2

88. Correct stability order is

- (1) $\text{F}^\ominus > \text{Cl}^\ominus > \text{Br}^\ominus > \text{I}^\ominus$
- (2) $\text{I}^\ominus > \text{F}^\ominus > \text{Br}^\ominus > \text{Cl}^\ominus$
- (3) $\text{I}^\ominus > \text{Br}^\ominus > \text{Cl}^\ominus > \text{F}^\ominus$
- (4) $\text{Br}^\ominus > \text{F}^\ominus > \text{Cl}^\ominus > \text{I}^\ominus$

89. Correct statement :



- (4) All of these

90. Match the column

Column - I		Column - II	
(a)		(p)	Number of chiral center = 1
(b)		(q)	Number of chiral center = 2
(c)		(r)	Number of chiral center = 3
(d)		(s)	Number of chiral center = 4

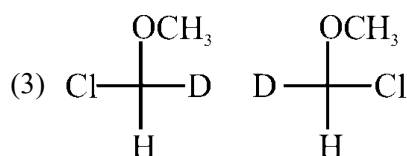
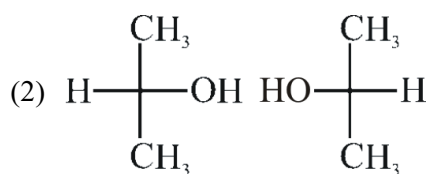
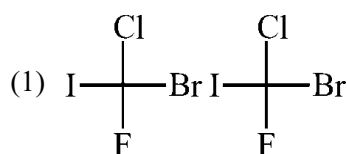
(1) $a \rightarrow p, b \rightarrow q, c \rightarrow r, d \rightarrow s$

(2) $a \rightarrow q, b \rightarrow r, c \rightarrow p, d \rightarrow s$

(3) $a \rightarrow p, b \rightarrow s, c \rightarrow r, d \rightarrow q$

(4) $a \rightarrow p, b \rightarrow q, c \rightarrow s, d \rightarrow r$

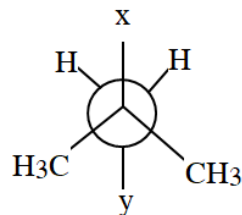
91. Which of the following represent pair of enantiomers ?



(4) None

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92. Identify X and Y for



(1) $-\text{CH}_3, -\text{CH}_3$

(2) $-\text{CH}_3, -\text{CH}_2-\text{CH}_3$

(3) $-\text{C}_2\text{H}_5, -\text{C}_2\text{H}_5$

(4) $-\text{CH}_3, -\text{H}$

93. For strong electrolytes, Λ_m increases slowly with dilution and can be represented by equation $\Lambda_m = \Lambda_m^0 - A\sqrt{C}$, Here value of constant A for a given solvent and temperature will be different for which electrolyte from rest -

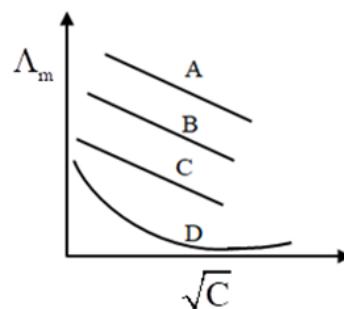
(1) CaCl_2

(2) MgCl_2

(3) MgSO_4

(4) $\text{Mg}(\text{NO}_3)_2$

94. Λ_m v/s $\sqrt{C}$ graph is plotted below for electrolytes A, B, C and D in aq. solutions -



then select the correct option, regarding these electrolytes -

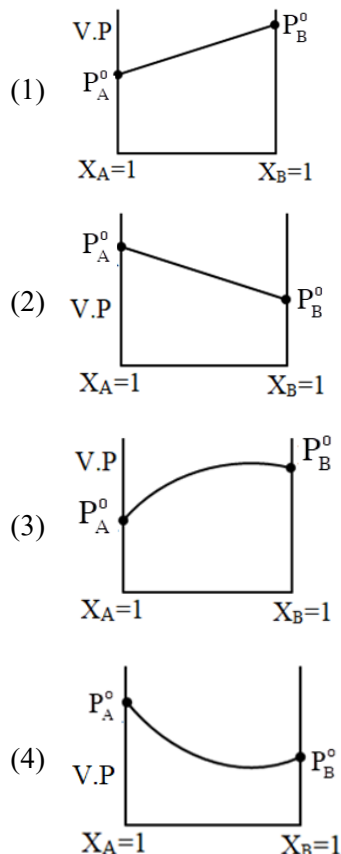
(1) $\text{A} = \text{HCl}, \text{B} = \text{NaCl}, \text{C} = \text{KCl}, \text{D} = \text{CH}_3\text{COOH}$

(2) $\text{A} = \text{KCl}, \text{B} = \text{NaCl}, \text{C} = \text{HCl}, \text{D} = \text{CH}_3\text{COOH}$

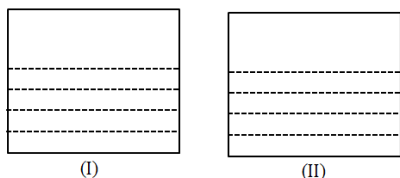
(3) $\text{A} = \text{HCl}, \text{B} = \text{KCl}, \text{C} = \text{NaCl}, \text{D} = \text{CH}_3\text{COOH}$

(4) $\text{A} = \text{HCl}, \text{B} = \text{CH}_3\text{COOH}, \text{C} = \text{KCl}, \text{D} = \text{NaCl}$

95. Boiling point of two liquids A and B are respectively 65°C and 80°C , then select the correct graph, Which represent vapour pressure of ideal solution formed by A and B with their composition in liquid solution -



96.



Both beaker (I) and (II) contains equal mass of water, In beaker (I), 10 g of non volatile solute A and in beaker II, 10 g of non volatile solute B is added, if molar mass of A is twice the molar mass of B and both solutes are non electrolytes, then

- (1) Freezing point of solution of A is more than freezing point of solution of B.
- (2) Boiling point of solution of A is more than boiling point of solution of B.
- (3) Freezing point of both solution is same
- (4) Osmotic pressure of both solution is same

at 300 K.

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97. Electrolytic conductance depends upon
- (1) Nature of electrolyte
 - (2) Viscosity of solvent
 - (3) Temperature
 - (4) All of the above
98. For a binary ideal liquid solution, the total pressure of the solution is given as
- (1) $P_{\text{total}} = P_A^0 + (P_A^0 - P_B^0)X_A$
 - (2) $P_{\text{total}} = P_B^0 + (P_A^0 - P_B^0)X_A$
 - (3) $P_{\text{total}} = P_A^0 + (P_B^0 - P_A^0)X_A$
 - (4) $P_{\text{total}} = P_B^0 + (P_B^0 - P_A^0)X_A$
99. An aqueous solution of 0.01 M KCl causes the same elevation in boiling point as an aqueous solution of urea causes the depression in freezing point. The concentration of urea solution is : ($K_b = 0.52$, $K_f = 1.86$) ($M = m$)
- (1) 0.01 M
 - (2) 0.005 M
 - (3) 0.02 M
 - (4) 0.04 M
100. The solubility of a specific non-volatile salt is 4 g in 100 g of water at 25°C . If 2.0 g, 4.0 g and 6.0 g of the salt added to 100 g of water at 25°C , in system X, Y and Z. The vapour pressure would be in the order :-
- (1) $X < Y < Z$
 - (2) $X > Y > Z$
 - (3) $Z > X = Y$
 - (4) $X > Y = Z$

SECTION-A (BIOLOGY-I)

101. Select the incorrect pair from following ?

- (1) Biosphere reserve - 115
- (2) National park - 90
- (3) Wild life sanctuaries - 448
- (4) Hot spot - 34

102. Which of the following is an insitu conservation method ?

- (1) National park (2) Tissue culture
- (3) Zoological park (4) Botanical garden

103. Chipko movement was launched for the protection of :-

- (1) livestock (2) wetlands
- (3) grasslands (4) forests

104. The Joint Forest Management (JFM) was introduced by the govt. of India in :-

- (1) West Bengal in 1950s
- (2) Madhya Pradesh in 1741s
- (3) Kerela in 1960s
- (4) West Bengal in 1980s

105. According to Robert May the global species diversity is about :-

- (1) 7 million (2) 12 million
- (3) 50 million (4) 20 million

106. "Within a region, species richness increases with increasing explored area, but only upto a limit. "This was the observation of :-

- (1) Paul Ehrlich
- (2) Robert May
- (3) Kofi Annan
- (4) Alexander von Humboldt

107. Which one of the following have the highest number of species in nature ?

- (1) Fungi (2) Insects
- (3) Birds (4) Angiosperms

108. In Rivet poper hypothesis 'rivet' represents :

- (1) Aeroplane (2) Ecosystem
- (3) Community (4) Species

109. Small animals are rarely found in polar regions. The reason is their :-

- (1) High cost of the thermoregulation with respect to energy consumption
- (2) Low body temperature
- (3) Low surface to volume ratio
- (4) High rate of respiration

110. Humus is degraded by some microbes and it leads to release of inorganic nutrients. This occurs by process known as ?

- (1) Decomposition (2) Humification
- (3) Mineralisation (4) Catabolism

111. Which of the following biome belongs to minimum - mean annual precipitation and minimum - mean annual temperature ?

- (1) Tropical forest
- (2) Coniferous forest
- (3) Grassland
- (4) Arctic and Alpine Tundra

112. Rate of release of nutrients into the atmosphere is regulated by which of the environmental factors?

- (1) Soil
- (2) Moisture
- (3) Temperature
- (4) All of the above

113. The components of the ecosystem are seen to function as a unit in following aspects except:-
 (1) Productivity (2) Energy Flow
 (3) Stratification (4) Decomposition
114. Organisms which are restricted to a narrow range of salinity are called ?
 (1) Euryhaline (2) Stenohaline
 (3) Stenothermal (4) Eurythermal
115. For which of the following PAR is not useful?
 (1) Plants
 (2) Photoautotrophic bacteria
 (3) Phytoplankton
 (4) Chemosynthetic bacteria
116. How many of the following are examples of aquatic ecosystem ?
 River, Wetland, Estuary, Desert, pond
 (1) 4 (2) 2 (3) 3 (4) 5
117. Which of the following is NOT an example of natural ecosystem ?
 (1) Grassland (2) Ocean
 (3) Crop field (4) Wetland
118. Amount of energy assimilated into biomass by consumers is known as ?
 (1) NPP
 (2) GPP
 (3) Secondary productivity
 (4) Net community productivity
119. Why is earthworm referred to as Farmer's Friend ?
 (1) It Feeds old tissues of crop, so that new tissues can form at the location of old tissues
 (2) Helps in absorption of minerals by crops
 (3) Feeds plant parasites that harm crops
 (4) Helps in fragmentation of detritus and in loosening of soil

120. Which factors inhibit decomposition process ?
 (1) Nitrogen and sugar rich Detritus
 (2) Warm and moist environment
 (3) Detritus rich in water content
 (4) Low temperature and Anaerobiosis
121. Plants capture _____ of photosynthetically active radiation (PAR). Fill in the blanks.
 (1) 1-5% (2) 2-10%
 (3) 5-10% (4) Less than 50%
122. **Assertion :-** Pyramid of energy is always upright, can never be inverted.
Reason :- When energy flows from a particular trophic level to the next trophic level, some energy is always lost as heat at each step.
 (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (3) Assertion is True but the Reason is False.
 (4) Both Assertion & Reason are False.
123. Under unfavourable conditions, many ____ (i) ____ species are known to enter ____ (ii) ____, a stage of suspended development. Fill in the blanks with appropriate option.
 (1) (i)-phytoplankton, (ii)-Diapause
 (2) (i)-Snails, (ii)-Aestivation
 (3) (i)-Bears, (ii)-Torpor
 (4) (i)-Zooplankton, (ii)-Diapause
124. Top soil of tropical rain forest is ?
 (1) Light coloured and thin
 (2) Dark coloured and thick
 (3) Dark coloured and thin
 (4) Light coloured and thick

125. Robert Constanza and his colleagues have very recently tried to put price tags on nature's life support services. With reference to this, select incorrect one.
- (1) Soil formation - US \$ 16.5 trillion
 - (2) Nutrient cycling - US \$ 3.3 trillion
 - (3) Climate regulation - US \$ 10 trillion
 - (4) Habitat for wild life - US \$ 1.98 trillion
126. Which of the following statement is incorrect ?
- (1) Energy at higher trophic levels is higher compared to lower trophic level
 - (2) A same species can occupy more than one trophic level
 - (3) In most ecosystems, all pyramids of number, energy and biomass are upright
 - (4) Trophic level represent functional level not a species as such
127. Next to temperature which of the following is an important abiotic factor ?
- (1) Light
 - (2) Topography
 - (3) Soil
 - (4) Water
128. Ecosystems need a constant supply of energy to synthesis the molecules as they require to :-
- (1) Increase the photosynthetic efficiency
 - (2) Reduce respiratory loss
 - (3) Counteract the universal tendency towards increasing disorderliness
 - (4) Control impact of predation
129. Which of the following is not an example of stenothermal organism ?
- (1) Mango tree
 - (2) Tuna fish
 - (3) Human
 - (4) Snow leopard
130. What is the salt concentration (measured as salinity in parts per thousand) in inland water ?
- (1) 10
 - (2) 0.5
 - (3) 30
 - (4) 5
131. Which of the following algae is found in deepest waters ?
- (1) Chemosynthetic autotrophs
 - (2) Red algae
 - (3) Brown algae
 - (4) Green algae
132. Which of the following is cues for timing foraging, migratory and reproductive activities of animals ?
- (1) Quality of light
 - (2) Variation in temperature
 - (3) Osmotic concentration of ambient environment
 - (4) Variation in light intensity and duration
133. If the stressful conditions are localised or remain only for a short duration, what are the alternatives that organism have to survive ?
- (1) Conform
 - (2) Migrate
 - (3) Suspend
 - (4) Either (2) or (3)
134. Desert lizards manage to keep their body temperature fairly constant by what means ?
- (1) Behavioural
 - (2) Morphological
 - (3) Physiological
 - (4) Biochemical

135. How many different genetical strains of Rice are present in India ?

- (1) 2,00,000
- (2) 50,000
- (3) 60,000
- (4) 1,00,000

SECTION-B (BIOLOGY-I)

136. Which statement is incorrect ?

- (1) Constant environment promote niche specialisation and lead to a greater species diversity
- (2) Rich biodiversity is not only essential for ecosystem health but imperative for the very survival of the human race on this planet
- (3) Species inventories are more complete in tropical than in temperate regions.
- (4) The number of fungi species in the world is more than the combined total of the species of fishes, amphibians, reptiles and mammals.

137. In species area relationship the relationship is represented as-

- (1) Rectangular parabola
- (2) Rectangular Hyperbola
- (3) Sigmoid
- (4) Straight line

138. Biodiversity term was popularised by :-

- (1) Odum
- (2) Robert May
- (3) Edward Wilson

(4) Tansley
TG: @Chalnaayaaar

139. Which of the following forest is called as 'lungs of planet'?

- (1) Temperate forest (2) Amazon forest
- (3) Tundra forest (4) Taiga forest

140. Which scientist proposed "Rivet popper hypothesis" related to biodiversity and ecosystem ?

- (1) David Tilman
- (2) Paul Ehrlich
- (3) Tansley
- (4) Warming

141. Which one of the following shows maximum genetic diversity in India ?

- (1) Groundnut (2) Rice
- (3) Maize (4) Mango

142. Biologists have not yet been able to estimate the prokaryotic species diversity as :-

- (1) Perfect sexual reproduction is common in prokaryotes
- (2) Conventional taxonomic methods are not suitable for identifying them
- (3) Most of the species can be easily grown in culture medium under laboratory conditions
- (4) Alternation of generation is common in prokaryotes

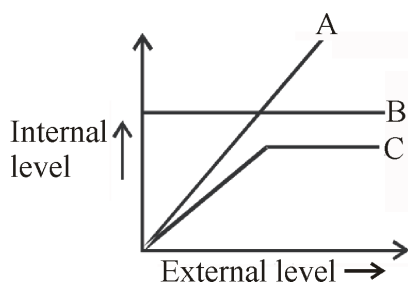
143. Producer Pri Sec Top

- (I) Plant → Insect → Frog → Eagle
- (II) Plant → Rat → Snake → Peacock

When food chain I will connected with food chain-II and snake eats frog then snake is :-

- (1) Primary consumer
- (2) Secondary consumer
- (3) Tertiary consumer
- (4) Top consumer

144.



Consider the above graph and choose the correct option :

	A	B	C
1	Maintains Homeostasis	Can not maintain Homeostasis	Maintains Homeostasis
2	Maintains Homeostasis upto a level	Maintain Homeostasis	Cannot maintain Homeostasis
3	Maintains Homeostasis	Maintain Homeostasis upto a level	Cannot maintain Homeostasis
4	Cannot maintain Homeostasis	Maintain Homeostasis	Maintain Homeostasis upto a level

145. What percent does the soil formation account for out of the total costs of various ecosystem services ?

- (1) 20 (2) 50 (3) 30 (4) 40

146. Match column-I with column-II and choose correct option ?

Column-I		Column-II	
(a)	Productivity	(i)	Input
(b)	Food chain/web	(ii)	Transfer of energy/Minerals
(c)	Nutrient cycling	(iii)	Output
(d)	Degradation and Energy loss		

- (1) a-(i), b-(i), c-(ii), d-(iii)
 (2) a-(ii), b-(ii), c-(i), d-(iii)
 (3) a-(ii), b-(ii), c-(ii), d-(iii)
 (4) a-(i), b-(ii), c-(ii), d-(iii)

TG: @Chalnaayaaar

147. What is the NPP of oceans ?

- (1) 55 million tons per year
 (2) 55 million tons per m² per year
 (3) 55 billion tons per year
 (4) 55 billion tons per m² per year

148. How can farmland be used to transfer the greatest amount of energy to humans ?

- (1) By keeping cows to provide milk and curd
 (2) By growing plants for people to eat
 (3) By keeping sheep to provide meat
 (4) By using half the land for crops and half for meat production

149. How many of the following statements are correct ?

- (A) NPP is available biomass for consumption to consumers
 (B) Ecological pyramid assumes simple food chain
 (C) In aquatic ecosystem GFC is major conduit of energy flow
 (D) DFC can be connected to GFC at some level

- (1) 3
 (2) 1
 (3) 4
 (4) 2

150. Atmospheric inputs of phosphorus through rainfall are much __ (i) __ than carbon inputs, and secondly, gaseous exchanges of phosphorus between organism and environment are __ (ii) __. Fill in the blanks with appropriate answer.

- (1) (i)-larger, (ii)-significant
 (2) (i)-smaller, (ii)-negligible
 (3) (i)-larger, (ii) negligible
 (4) (i)-smaller, (ii)-significant

SECTION-A (BIOLOGY-II)

151. Which of the following is not vestigial organ in humans :-

- (1) Ear pinna (2) Liver
(3) Coccyx (4) both (1) and (2)

152. Which is the correct pairs of animals showing adaptive radiation : —

- (1) Tasmanian tiger cat, flying phalanger
(2) Bobcat, Wombat
(3) Kangaroo, placental mouse
(4) Koala, placental wolf

153. Mark the option with the correct pair of Australian marsupials (from column-I) and placental mammals (from column-II) showing convergent evolution?

	Column-I		Column-II
A	Numbat	i	Anteater
B	Spotted cuscus	ii	Flying squirrel
C	Sugar glider	iii	Lemur
D	Tiger cat	iv	Mole
		v	Bobcat

- (1) A-i; B-v; C-ii; D-iii
(2) A-i; B-iii; C-ii; D-v
(3) A-iv; B-iii; C-ii; D-v
(4) A-i; B-v; C-i; D-iv

154. The process by which organisms with common evolutionary history evolve different phenotypic adaptations in response to different environmental challenges, is called-

- (1) Adaptive radiation
(2) Artificial selection
(3) convergent evolution
(4) Non random evolution

TG: @Chalnaayaaar

155. According to Oparin-Haldane theory :-

- (1) Life originated from pre-existing life
(2) Primitive atmosphere was highly oxidising
(3) First organic molecule formed was methane
(4) First form of life were chemoautotrophs

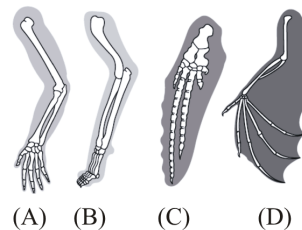
156. Among the following structures, which is homologous to the wing of a bird ?

- (1) Dorsal fin of a shark
(2) Hindlimb of a kangaroo
(3) Caudal fin of a flying fish
(4) Flipper of an aquatic mammal

157. During evolution, the first living being were prokaryotic cell, like

- (1) Bacteria (2) Virus
(3) Blue green algae (4) Actinomycetes

158. The four sketches (A, B, C and D) given below represent four different types of fore-limbs of mammals. Which one of these is correctly identified in the option given along with its correct function and adaption.



		Fore limb	Function	Adaptation
(1)	(D)	Bat	Flying	Aerial
(2)	(A)	Man	Running	Carnivore
(3)	(B)	Whale	Swimming	Aquatic
(4)	(C)	Cheetah	Running	Carnivore

159. Golden age of amphibians

- (1) Jurassic (2) Carboniferous
(3) Devonian (4) Triassic

160. A process in which heritable variations enabling better survival are enabled to reproduce and leave greater number of progeny is called :-
(1) Genetic drift (2) Natural selection
(3) Founder effect (4) Both (1) and (3)
161. An important evidence in favour of organic evolution is the occurrence of –
(1) Homologous and vestigial organs
(2) Analogous and vestigial organs
(3) Homologous organs only
(4) Homologous and analogous organs
162. In evolution the studies can be made at molecular level. For example the protein present in the blood of man and ape are similar. The base sequence in nucleic acids and amino acids sequence in protein in related organism is alike. These are the examples which one specifically referred to in :-
(1) convergent evolution
(2) molecular analogy
(3) molecular homology
(4) homoplastic appearance
163. **Assertion** :- All mammals like whales, bats, cheetah and human etc. share similarities in the pattern of bones of forelimbs.
Reason :- Eutherians show good example of divergent evolution.
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are true but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.
164. **Assertion** : Amphibians evolved into reptiles which are first true terrestrial vertebrates.
Reason : Reptiles lay thick shelled eggs which do not dry up in sun.
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion.
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion.
(3) Assertion is true statement but Reason is false.
(4) Both Assertion and Reason are false statements.
165. **Statement-I** : Coelacanth was synapsid.
Statement-II : Coelacanth arose in 1938.
(1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & Statement-II are incorrect.
166. **Statement-I** : When we look at the stars at clear night sky we are in the way looking back in time.
Statement-II : Stellar distances are measured in light years.
(1) Statement I and II both are correct
(2) Statement I and II both are incorrect
(3) Only statement I is correct
(4) Only statement II is correct
167. Which one of the following is a common ancestor of birds, mammals and crocodiles :-
(1) Sauropsids (2) Synapsids
(3) Early reptiles (4) Pelycosaurs

168. If more than one adaptive-radiation appeared to have occurred in an isolated geographical area, one can call this :-
- (1) Co-evolution
 - (2) Divergent evolution
 - (3) Convergent evolution
 - (4) Homology
169. In the solar system of the milky way galaxy earth was supposed to have been formed about how many years after the formation of universe ?
- (1) 4.5 billion years back
 - (2) 4.5 million years back
 - (3) After 15.5 billion years
 - (4) After 5 million years
170. Which of the following statement is correct ?
- (1) Earth is very old almost 20 billion year old.
 - (2) Huge clusters of galaxies comprise the earth.
 - (3) Considering the size of universe, earth is indeed a speck.
 - (4) Big bang talks of a plural small explosion.
171. Which theory for origin of life is still a favourite idea for few astronomers ?
- (1) Special creation
 - (2) Cosmic panspermia
 - (3) Autogenesis
 - (4) Oparin-Haldane Theory
172. Which of the following is the reptilian ancestor of mammals ?
- (1) Sauropsids
 - (2) Synapsids
 - (3) Thecodonts
 - (4) Chlrophyta
173. Darwin's finches provide excellent evidence in favour of evolution through this concept we can understand concept of :-
- (1) Comparative anatomy
 - (2) Adaptive radiation
 - (3) Embryology
 - (4) Paleontology
174. Given below are two statements : one is labelled as assertion (A) and the other is labelled as reason (R).
- Assertion (A) :** Wings of birds and wings of butterfly are an example of analogous organs.
- Reason (R) :** Wings of birds and wings of butterfly have different origin but same functions.
- In the light of the above statement, choose the most appropriate answer from the options given below :
- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 - (2) (A) is correct but (R) is not correct
 - (3) (A) is not correct but (R) is correct
 - (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)
175. Key concepts of Darwinian theory of evolution are ?
- (1) Branching descent only
 - (2) Natural selection only
 - (3) Saltation
 - (4) Branching descent and natural selection
176. Select the analogous structure :-
- (1) Wings of butterfly and wings of birds.
 - (2) Flippers of dolphin and leg of horse.
 - (3) Reddish and carrots
 - (4) Mouth part of insecta

177. 'Ontogeny recapitulates phylogeny' is explained by :-

- (1) Theory of natural selection
- (2) Recapitulation theory
- (3) Mutation theory
- (4) Theory of pangenesis

178. Match the columns :-

	Column-I		Column-II
(A)	Lamarck	(i)	Theory of inheritance of acquired character
(B)	Weismann	(ii)	Theory of germplasm
(C)	Darwin	(iii)	Descent with modification
(D)	Hugo de Vries	(iv)	Mutation theory

- (1) A-ii, B-i, C-iii, D-iv
- (2) A-i, B-ii, C-iv, D-iii
- (3) A-i, B-iv, C-iii, D-ii
- (4) A-i, B-ii, C-iii, D-iv

179. Dinosaurs suddenly disappeared from the earth :-

- (1) 200 mya
- (2) 65 mya
- (3) 25 mya
- (4) 500 mya

180. **Statement-I** : The origin of life is considered a unique event in the history of universe.

Statement-II : The gases condensed under gravitation are formed galaxies.

Choose the correct answer from the given option :-

- (1) Both statement-I and statement-II are incorrect
- (2) Both statement-I and statement-II are correct
- (3) Statement-I is correct but statement-II is not correct
- (4) Statement-I is incorrect but statement II is correct

correct
TG: @Chalnaayaaar

181. Which of the following fell to form coal deposits ?

- (1) Cycads
- (2) Gnetales
- (3) Conifers
- (4) Giant ferns

182. Which of the following was about 20 feet in height and had huge fearsome dagger like teeth ?

- (1) Tyrannosaurus
- (2) Branchiosaurus
- (3) Stegosaurus
- (4) Triceratops

183. Identify the correct match from following ?

	Column-A	Column-B
(1)	Invertebrate	300 MYA
(2)	Jawless fish	350 MYA
(3)	Sea weeds	500 MYA
(4)	Coelacanth	320 MYA

184. How many of the following are homologous structure in plants ?

- (a) Potato and ginger
- (b) Radish and carrot
- (c) Testes in male and ovaries in female
- (d) Sting of bee and scorpion
- (e) Thorn of bougainvillea and tendril of Cucurbita
- (f) Sweet potato and potato

- (1) Four
- (2) Three
- (3) Two
- (4) Five

185. When two species of different genealogy come to resemble each other as result of adaptation, the phenomenon is termed as –

- (1) Convergent evolution
- (2) Divergent evolution
- (3) Micro evolution
- (4) Co-evolution

SECTION-B (BIOLOGY-II)

186. "Snakes lost their legs" this examples was given in which theory ?

- (1) Lamarckism (2) Darwinism
(3) Neodarwinism (4) Mendelism

187. Earliest fossil form in the phylogeny of horse is :-

- (1) Equus (2) Meshippus
(3) Eohippus (4) Merychippus

188. Which one of the following options gives one correct example each of convergent evolution and divergent evolution-

	Convergent evolution	Divergent evolution
(1)	Eye of octopus and mammal	Bones of forelimb of vertebrates
(2)	Bones of forelimb of vertebrates	Wings of butterfly and birds
(3)	Eye of octopus and mammal	Wings of butterfly and birds
(4)	Thorn of <i>Bougainvillea</i> and tendril of <i>Cucurbita</i>	Eye of octopus and mammal

189. Which is the most important factor for evolution of new species :

- (1) Mimicry
(2) Extensive inbreeding
(3) Reproductive isolation
(4) Adaptation

190. Which statement is wrong for the concept of natural selection ?

- (1) Fitness is not required
(2) Pre-adaptive advantageous mutations results into new phenotypes
(3) Nature selects for the fitness
(4) No variant is completely wiped out by the nature

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191. One of the revolutionary concepts in biology was Charles Darwin's "Origin of Species" it deals with:-

- (1) Gene mutation
(2) Use and disuse of organs
(3) Germplasm theory
(4) Natural selection leading to survival of fittest

192. Which of the following is **NOT** a component of the theory of evolution by natural selection?

- (1) Competition for food and space
(2) Variation among species
(3) Inheritance of acquired characteristics
(4) Survival and reproduction

193. **Assertion :-** Due to continental drift pouched mammals of Australia survived.

Reason :- There is lack of competition from any other mammal.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

194. **Statement-I :** Single step large mutation is saltation.

Statement-II : Saltation was the idea of Darwin.

- (1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & Statement-II is incorrect.

195. The main reason why antibiotics could not solve all the problems of bacteria mediated diseases is

- (1) insensitivity of the individual following prolonged exposure to antibiotics
- (2) inactivation of antibiotics by bacterial enzymes
- (3) decreased efficiency of the immune system
- (4) the development of mutant strains resistant to antibiotics

196. Select the correct statement :-

- (1) Fitness is the end result of the ability to adapt and get select by nature
- (2) Natural selection and genetic variation are two main key point of Darwinism theory of evolution
- (3) Adaptive ability is always inherited
- (4) Placental wolf and Tasmanian wolf are example of homology

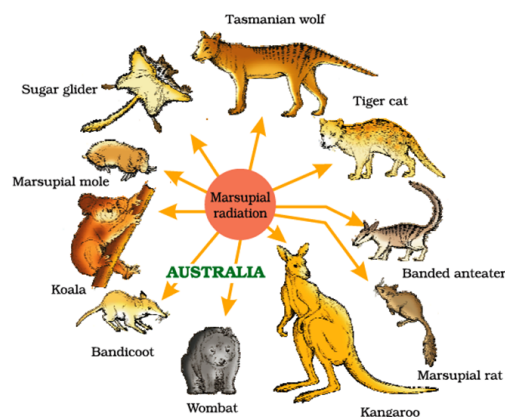
197. According to Hugo de vires, mutation causes speciation and a single step large mutation is called as :-

- (1) Genetic drift (2) Saltation
- (3) Adaption (4) Isolation

198. Darwin Judged the fitness of individual through :-

- (1) Ability to defend
- (2) Strategy for obtaining food
- (3) Number of offspring
- (4) Dominance over others

199. How many statements are correct about the diagram given below ?



- (a) Australian marsupials
- (b) Shows homology
- (c) Example of divergent evolution
- (d) All are examples of adaptive radiation

- (1) One
- (2) Two
- (3) Three
- (4) Four

200. Suppression of South America Fauna (like hippopotamus, bear, rabbit etc.) by North American Fauna can be best explained by :-

- (1) Atavism
- (2) Biogeographical evidences
- (3) Embryological evidences
- (4) Palaentological



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FROM CURRENT & PREVIOUS UNITS

(Session : 2023 - 2024)

ENTHUSE ADV. COURSE : MEC

MINOR TEST NO. & DATE	CURRENT UNIT	PREVIOUS UNIT (COMPLETE TOPIC)	
03 21-05-2023	SYLLABUS COVERED FROM 24-04-2023 TO 20-05-2023	P	NLM & FRICTION AND GRAVITATION
		C	CHEMICAL KINETICS
		B	HUMAN HEALTH & DISEASE

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